

Analysis of Propeller Aircraft Performance on Titan's Atmosphere Using Differential Equations

May 2024

19 pages

IB PERSONAL NUMBER: knw165

We are living in an age where an entirely new frontier is opening up to humanity, giving us a chance to innovate and expand which we have not seen since the discovery of the New World centuries ago. I am, of course, referring to the breakneck pace in which humanity is venturing into outer space, being able to barely lift off the ground in the early 1900s to landing a man on the Moon 70 years later and establishing a permanent presence in the International Space Station almost 100 years later. The human race's thirst to conquer all challenges which it faces has led to technological advancements which has slowly turned the realm of science fiction into just science. I still remember vividly watching anxiously NASA's livestream as nervous scientists huddled around a screen to see if *Perseverance*, the newest rover to roam on Mars, would successfully land on the Martian surface. As the scientists celebrated when they got confirmation that it landed successfully, all I could think about was the package inside *Perseverance*: *Ingenuity*, a small two-blade helicopter, which, if successful, would become the first powered flight on another planet. On April 19th, 2021, this became a reality, and by the time of writing this paper *Ingenuity* has flown for over two hours and flown a total distance of 10 miles, symbolizing a triumph of engineering which factored Mars' thin atmosphere, its lower gravity, and the necessity to keep the helicopter lightweight ("Mars Helicopter").

Watching *Ingenuity* take flight sparked an interest in outer space and the development of aircraft on other planets, which is why I became ecstatic in discovering that NASA plans to land a drone on another body by 2034 – *Dragonfly*. It will fly across Titan, a moon of Saturn, which is unique in it having an atmosphere four times denser than Earth, all while being exposed to -179-degree temperatures (Hautaluoma and Johnson). Exploring Titan is critical to understanding the origin of life on Earth due to the discovery of liquid water underneath its surface and a complex chemical environment which could harbor life ("Titan: Facts").

Inspired by the success of *Ingenuity* and the future plans for *Dragonfly*, I wish to explore further the physical factors at play in the development and designing of aircraft meant to fly on other planets. As a result, this essay will have the aim of simulating how a Cessna 172, the most mass-produced aircraft in history, would fair on Titan's surface and atmosphere, taking into consideration Titan's reduced gravity, higher air density, as well as the specifications of the Cessna 172's known dimensions and performance. Through this essay, I aim to understand better the fundamental forces at play in aviation and gain an insight into how engineers and designers come up with their models.

In this essay, I will go over the basic forces at play in aircraft, as well as equations for the specific thrust of a propeller, to determine several key qualities of the Cessna 172 which are necessary for flight, like its coefficient of lift and safe rotational speed of the propellers due to a high Reynolds Number. Upon doing so, I will analyze how the Cessna would perform in Titan's atmosphere, including top speed, minimum speed, maximum carrying capacity, optimal amount of cargo, and the range of the aircraft. All these will be done through differential equations until we can get a function with mass as an input and distance as an output. In conclusion, this paper will go over several key aspects of flight and imagine a futuristic world where powered manned flight on another world is possible.

The Cessna 172 is the most successful aircraft in history, selling more than 45,000 planes to experienced and new pilots alike due to how easy it is to pilot (O'Hare). As a result, a comparable design would be of interest for astronauts with flight experience but who are in a new to a foreign atmosphere with different characteristics. First, though, it is necessary to examine the basic characteristics of the Cessna 172 on Earth to then evaluate its performance on Titan. On an aircraft, there are four basic forces acting on it during flight: Its drag, caused by air

resistance as it moves through the air, its thrust, which opposes and overcomes the drag and is provided by the engines of the plane, the weight of gravity pulling down on the plane, and the lift generated by the wings which counteract gravity (Hall, “Four Forces on an Airplane”).

We can, therefore, begin with the lift of the Cessna 172. The lift of an aircraft can be determined by the equation:

$$L = \frac{1}{2} \rho v_{\infty}^2 C_L S_{ref}$$

Where L is the lift produced in Newtons (N), ρ is the air density in kilograms per cubic meter ($\frac{kg}{m^3}$), v_{∞} is the velocity of the plane against the air flow in meters per second (m/s), C_L is the coefficient of lift of the wings, a unitless measure of how efficiently the lift is produced, and S_{ref} is the surface area of the surface producing lift, in this case the wings, in square meters (m^2) (Burgers 1). All of these values are easily findable except for the coefficient of lift. However, we can determine this from the airplane’s stall speed, or the minimum speed necessary for the aircraft to stay aloft using its own lift. In other words, at its stall speed, the lift is exactly equal to its weight. In the Cessna 172, it is 89 kilometers/hour, or $24.7\bar{2}$ m/s. Considering that the formula for weight due to gravity is $F = mg$, where m is the aircraft’s mass and g is the acceleration due to gravity, $9.8 \frac{m}{s^2}$, that the Cessna 172 has a wing surface area of 174 square feet, or $16.165 m^2$, that air density at sea level on Earth is $1.293 \frac{kg}{m^3}$, and that the Cessna has a maximum takeoff weight of 1,157 kg, we can determine its coefficient of lift (“Air Mass/Density”; “Cessna Skyhawk”; Hall, “Weight Equation”).

$$Lift = Weight \rightarrow \frac{1}{2} \rho v_{\infty}^2 C_L S_{ref} = mg$$

$$C_L = \frac{2mg}{\rho v_{\infty}^2 S_{ref}} = \frac{2(1157kg) \left(9.8 \frac{m}{s^2}\right)}{\left(1.293 \frac{kg}{m^3}\right) \left(24.72 \frac{m}{s}\right)^2 (16.165 m^2)} \approx 1.775$$

Having found the Cessna's coefficient of lift, we can move onto another important characteristic of flight: turbulent flow. When an aircraft's propeller flows too fast through the air, turbulent air flow can occur which destabilizes the aircraft and reduces its efficiency (Draley 68). The Cessna 172 is advised to not exceed 302 km/h, or $83. \bar{8} \frac{m}{s}$ ("Cessna Skyhawk"). Additionally, the blades on the Cessna 172 can spin at most at 2700 rpm (McIver 2).

Having found this, the turbulent flow can be measured through the Reynolds Number, a unitless value representing how turbulent the movement of air through a moving object is ("Reynolds Number"). The Reynolds Number of a propeller can be found through this equation:

$$Re_b = \frac{\rho v_{\infty} c}{\mu} \sqrt{1 + \frac{1}{3} \left(\frac{\pi n d}{v_{\infty}} \right)^2}$$

Where c is the chord length, or $0.577 * radius_{propeller}$, μ is the dynamic viscosity coefficient, a measure of how "sticky" and resistant the air is, which is $1.7332 * 10^{-5} \frac{Ns}{m^2}$ in Earth's air, n is the revolutions per second of the blade, and d is the propeller diameter (Burgers 8,10,20; "Reynolds Number"). The Cessna 172 has a propeller diameter of 75 inches, or 1.905 meters, meaning $c = (0.577) \left(\frac{1.905m}{2} \right) = 0.550 m$ (McIver 7). To find the revolutions per second, we can take the fact that we know the top speed of 2700 rpm:

$$2700 rpm * \frac{1 \frac{rotation}{second}}{60 rpm} = 45 \frac{rotation}{second}$$

Finally, we plug in all variables into the equation,

$$\begin{aligned}
Re_b &= \frac{\rho v_{\infty} c}{\mu} \sqrt{1 + \frac{1}{3} \left(\frac{\pi n d}{v_{\infty}} \right)^2} \\
&= \frac{\left(1.293 \frac{kg}{m^3} \right) \left(83.8 \frac{m}{s} \right) (0.55 \text{ m})}{1.7332 * 10^{-5} \frac{Ns}{m^2}} \sqrt{1 + \frac{1}{3} \left(\frac{\pi \left(45 \frac{rev}{s} \right) (1.905 \text{ m})}{\left(83.8 \frac{m}{s} \right)} \right)^2} \\
&= 7,243,764.132
\end{aligned}$$

As a result, when applying this aircraft to the atmospheric conditions at Titan, we must attempt to try and keep it under a Reynolds Number of around $7.2 * 10^6$. Plugging back in the equation,

$$Re_b = \frac{\rho v_{\infty} c}{\mu} \sqrt{1 + \frac{1}{3} \left(\frac{\pi n d}{v_{\infty}} \right)^2} \quad n = \frac{1}{\frac{d\pi}{v_{tip}}} = \frac{v_{tip}}{d\pi}$$

As a result, we must find an equation for the velocity at the tip of the blades. To do so, we can equate the thrust of the aircraft provided by the propellers with the drag provided by the air, as when they are equal the plane is flying at constant speed. Now, one must use the drag equation:

$$Drag = \frac{C_d \rho v_{\infty}^2 S_{ref}}{2}$$

Notably, this utilizes the same variables as the lift equation except for C_d , the unitless drag coefficient which represents how efficiently the aircraft can push air out of the way (Hall, “Modern Drag Equation”). In a Cessna 172, this value stands at 0.032 (McIver 3). Meanwhile, the thrust equation for propellers is as follows:

$$T = \left(\frac{1}{4} v_{\infty}^2 + \frac{1}{6} v_{tip}^2 \right) \rho C_{L_b} S_b$$

Where v_{tip} is the velocity of the tip of the blade as it rotates in meters per second, S_b is the surface area of the blades, and C_{L_b} is the coefficient of lift of the blades (Burgers 1). Now, equating the two:

$$Thrust = Drag \rightarrow \left(\frac{1}{4} v_{\infty}^2 + \frac{1}{6} v_{tip}^2 \right) \rho C_{L_b} S_b = \frac{C_d \rho v_{\infty}^2 S_{ref}}{2}$$

$$\frac{1}{6} v_{tip}^2 = \frac{C_d v_{\infty}^2 S_{ref}}{2 C_{L_b} S_b} - \frac{1}{4} v_{\infty}^2$$

$$v_{tip} = \sqrt{6 \left(\frac{C_d v_{\infty}^2 S_{ref}}{2 C_{L_b} S_b} - \frac{1}{4} v_{\infty}^2 \right)}$$

Now we must find some of the unknowns, which are the coefficient of lift of the blades and their surface area.



Using the above image and Photoshop pixel measurement software, one can find that the wingspan of the image is 1531 pixels (“Cessna Skyhawk”). Meanwhile, each propeller blade was 106 pixels tall and 19 pixels wide. Since the wingspan of the Cessna 172 is 11 meters, that gives a measurement scale of 0.00718 meters/pixel, giving the blades measurements of 0.762 meters

long and 0.137 meters wide (“Cessna Skyhawk”). Assuming a rectangular-like shape and multiplying by two to account for two blades, that gives a total area of 0.208 square meters.

Meanwhile, to find the coefficient of lift of the blades, we can use the fact that the Cessna 172’s maximum cruising speed is 230 km/h, or 63.89 m/s (“Cessna Skyhawk”). This means that at this speed, the propeller’s maximum output thrust equals the resistance provided by drag. We can once again equate thrust and drag to find the coefficient of lift of the blades:

$$\begin{aligned}
 Thrust &= Drag \rightarrow \left(\frac{1}{4} v_{\infty}^2 + \frac{1}{6} v_{tip}^2 \right) \rho C_{L_b} S_b = \frac{C_d \rho v_{\infty}^2 S_{ref}}{2} \\
 C_{L_b} &= \frac{C_d \rho v_{\infty}^2 S_{ref}}{2 \left(\frac{1}{4} v_{\infty}^2 + \frac{1}{6} v_{tip}^2 \right) \rho S_b} = \frac{(0.032) \left(1.293 \frac{kg}{m^3} \right) \left(63.89 \frac{m}{s} \right)^2 (16.165 m^2)}{2 \left(\frac{1}{4} \left(63.89 \frac{m}{s} \right)^2 + \frac{1}{6} \left(85.725 \frac{m}{s} \right)^2 \right) \left(1.293 \frac{kg}{m^3} \right) (0.208 m^2)} \\
 &= 2.261
 \end{aligned}$$

Finally, with all of the unknown variables found, we can plug them all back into the Reynolds Number equation,

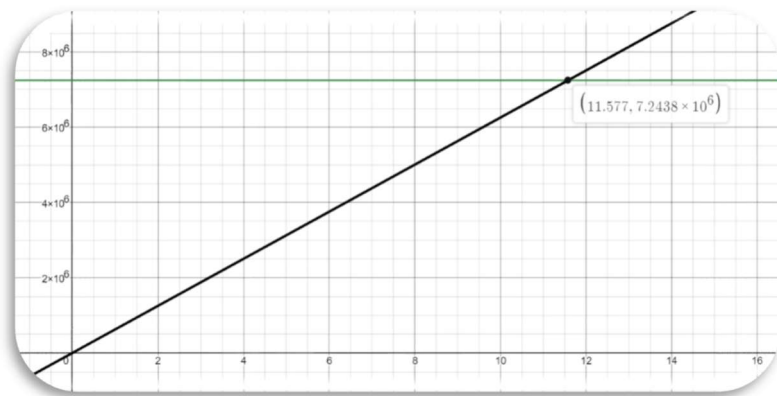
$$Re_b = \frac{\rho v_{\infty} c}{\mu} \sqrt{1 + \frac{1}{3} \left(\frac{\pi n d}{v_{\infty}} \right)^2} \quad n = \frac{1}{\frac{d\pi}{v_{tip}}} = \frac{v_{tip}}{d\pi} \quad v_{tip} = \sqrt{6 \left(\frac{C_d v_{\infty}^2 S_{ref}}{2 C_{L_b} S_b} - \frac{1}{4} v_{\infty}^2 \right)}$$

We can then re-write the Reynolds Number as a function of the forward velocity of the plane:

$$Re_b = \frac{\rho v_{\infty} c}{\mu} \sqrt{1 + \frac{1}{3} \left(\frac{\sqrt{6 \left(\frac{C_d v_{\infty}^2 S_{ref}}{2 C_{L_b} S_b} - \frac{1}{4} v_{\infty}^2 \right)}}{v_{\infty}} \right)^2}$$

Unfortunately, Titan has a higher atmospheric density of $5.4 \frac{kg}{m^3}$ (about 4 times that of Earth's) and a lower atmospheric viscosity $6 * 10^{-6} \frac{Ns}{m^2}$ (about 3 times smaller than Earth's), meaning there's more air molecules more prone to move around and interfere with the air wing, which only increases the Reynold's number (Lorenz et al 375). As a result, the maximum tolerance of the propellers will be much less than that of the on Earth.

Plotting this equation into Desmos' graphing calculator, with the Reynolds Number on the y-axis and the forward velocity as the x-axis:



The horizontal line is the limit for the Reynolds Number as defined above for safe flight for the Cessna 172. As seen, the number increases exponentially faster than on Earth, only allowing for a maximum horizontal speed of **11.577 meters per second**, or 25.9 miles per hour, a pitifully slow speed. At this speed, the tips of the propellers would be travelling at

$$v_{tip} = \sqrt{6 \left(\frac{C_d v_{\infty}^2 S_{ref}}{2 C_{L_b} S_b} - \frac{1}{4} v_{\infty}^2 \right)}$$

$$= \sqrt{6 \left(\frac{(0.032) \left(11.577 \frac{m}{s} \right)^2 (16.165 m^2)}{2(2.261)(0.208 m^2)} - \frac{1}{4} \left(11.577 \frac{m}{s} \right)^2 \right)} = 15.534 \frac{m}{s}$$

It is also important to see what the minimum speed the Cessna would be able to fly at is to find a range of flight speed, which can be found by equating the lift of the plane with its weight, since it will find the minimum speed to support its weight. For this case, we shall use the Cessna 172's minimum weight, 782 kilograms ("Cessna Skyhawk").

$$Lift = Weight \rightarrow \frac{1}{2} \rho v_{\infty}^2 C_L S_{ref} = mg$$

$$v_{\infty} = \sqrt{\frac{2mg}{\rho C_L S_{ref}}} = \sqrt{\frac{2(762 \text{ kg}) \left(1.35 \frac{m}{s^2}\right)}{\left(5.4 \frac{kg}{m^3}\right) (1.775)(16.165 \text{ m}^2)}} \approx 3.644 \frac{m}{s} = 8.151 \text{ miles per hour}$$

As a result, the Cessna 172 would have a worryingly small range of operation of only $11.577 \frac{m}{s} - 3.644 \frac{m}{s} = 7.933$ meters per second, requiring maximum concentration for the pilots to control between slowly falling off the sky and extreme instability of the flight.

As a result, it must be asked how much weight the Cessna could even handle taking off with. Once again equating lift to weight, and assuming the top speed the plane can achieve (11.577 m/s) without the propellers facing serious damage,

$$Lift = Weight \rightarrow \frac{1}{2} \rho v_{\infty}^2 C_L S_{ref} = mg$$

$$m = \frac{\rho v_{\infty}^2 C_L S_{ref}}{2g} = \frac{\left(5.4 \frac{kg}{m^3}\right) \left(11.577 \frac{m}{s}\right)^2 (1.775)(16.165 \text{ m}^2)}{2 \left(1.35 \frac{m}{s^2}\right)} = 7691.235 \text{ kg}$$

This means that the Cessna 172 can carry, at most, 7691.235 kg of fuel, cargo, and plane. This is more than six times the Cessna's maximum takeoff weight on Earth at 1,157 kg, demonstrating how useful Titan's low gravity and high atmospheric density is ("Cessna Skyhawk"). Luckily, the Cessna 172 can carry 201 liters of fuel which takes up 144 kilograms,

or 318 pounds, meaning we are free to explore, assuming a 70 kg pilot and 762 kg plane,
 $7691.235 \text{ kg} - 144 \text{ kg} - 70 \text{ kg} - 762 \text{ kg} = 6715.235 \text{ kg}$ of cargo for the performance of the
 Cessna in all weight compositions to maximize the distance we can travel (“Cessna Skyhawk”).
 The Cessna 172’s engine, the Lycoming IO-360, has a maximum 180 horsepower, but the slow
 rotational speeds mean the blades are more than likely going to need much less (which will affect
 the fuel consumption), so using the minimum fuel consumption listed, we find a consumption
 rate of 35 pounds per hour (“Operator’s Manual Lycoming” 324).

We can convert these to metric, with

$$35 \frac{\text{pounds}}{\text{hour}} * \frac{1 \text{ kg}}{2.205 \text{ pounds}} * \frac{1 \text{ hour}}{3600 \text{ s}} = 0.004409 \frac{\text{kg}}{\text{s}}$$

As one would with a flight on another planet, we are looking to maximize the possible
 distance travelled. As a result, we must derive an equation for the distance possible by using
 differential equations. First, we must recognize that the force propelling the plane forward is the
 difference between the forward propulsion of the propellers and the backwards force of drag. In
 other words,

$$\text{Force} = \text{Thrust} - \text{Drag} = \left(\frac{1}{4} v_{\infty}^2 + \frac{1}{6} v_{tip}^2 \right) \rho C_{L_b} S_b - \frac{C_d \rho v_{\infty}^2 S_{ref}}{2}$$

Force = mass * acceleration, which means we can determine the acceleration of the aircraft at
 any point by dividing by the total mass, or $m_{cargo} + m_{plane}$ (Tavernier):

$$a = \frac{\left(\frac{1}{4} v_{\infty}^2 + \frac{1}{6} v_{tip}^2 \right) \rho C_{L_b} S_b}{m_{cargo} + m_{plane}} - \frac{C_d \rho v_{\infty}^2 S_{ref}}{2(m_{cargo} + m_{plane})}$$

$$a = \frac{\rho C_{L_b} S_b}{4(m_{cargo} + m_{plane})} v_{\infty}^2 + \frac{\rho C_{L_b} S_b}{6(m_{cargo} + m_{plane})} v_{tip}^2 - \frac{C_d \rho v_{\infty}^2 S_{ref}}{2(m_{cargo} + m_{plane})}$$

$$a = \left(\frac{\rho C_{L_b} S_b}{4(m_{cargo} + m_{plane})} - \frac{C_d \rho S_{ref}}{2(m_{cargo} + m_{plane})} \right) v_{\infty}^2 + \frac{\rho C_{L_b} S_b}{6(m_{cargo} + m_{plane})} v_{tip}^2$$

$$a = \frac{\rho C_{L_b} S_b - 2C_d \rho S_{ref}}{4(m_{cargo} + m_{plane})} v_{\infty}^2 + \frac{\rho C_{L_b} S_b v_{tip}^2}{6(m_{cargo} + m_{plane})}$$

Acceleration, by definition, is the instantaneous velocity, meaning it can be represented as the derivative of velocity with respect to time,

$$\frac{dv}{dt} = \frac{\rho C_{L_b} S_b - 2C_d \rho S_{ref}}{4(m_{cargo} + m_{plane})} v_{\infty}^2 + \frac{\rho C_{L_b} S_b v_{tip}^2}{6(m_{cargo} + m_{plane})}$$

To simplify, $\frac{\rho C_{L_b} S_b - 2C_d \rho S_{ref}}{4(m_{cargo} + m_{plane})} = a$ and $-\frac{\rho C_{L_b} S_b v_{tip}^2}{6(m_{cargo} + m_{plane})} = b$:

$$\frac{dv}{dt} = av^2 - b$$

$$\frac{dv}{av^2 - b} = dt$$

$$\int \frac{dv}{av^2 - b} = \int dt$$

$$\int \frac{dv}{(\sqrt{a}v + \sqrt{b})(\sqrt{a}v - \sqrt{b})} = t + c$$

$$\int \left(\frac{A}{\sqrt{a}v + \sqrt{b}} + \frac{B}{\sqrt{a}v - \sqrt{b}} \right) dv = t + c$$

$$\rightarrow A(\sqrt{a}v - \sqrt{b}) + B(\sqrt{a}v + \sqrt{b}) = 1$$

$$\text{When } v = \sqrt{\frac{b}{a}}, \rightarrow A(0) + 2\sqrt{b}B = 1 \rightarrow B = \frac{1}{2\sqrt{b}}$$

$$\text{When } v = -\sqrt{\frac{b}{a}} \rightarrow 2\sqrt{b}A + B(0) = 1 \rightarrow A = -\frac{1}{2\sqrt{b}}$$

$$\int \left(-\frac{1}{2\sqrt{b}(\sqrt{av} + \sqrt{b})} + \frac{1}{2\sqrt{b}(\sqrt{av} - \sqrt{b})} \right) dv = t + c$$

$$\frac{1}{2\sqrt{b}} \left(\int \frac{dv}{\sqrt{av} - \sqrt{b}} - \int \frac{dv}{\sqrt{av} + \sqrt{b}} \right) = t + c$$

$$u = \sqrt{av} - \sqrt{b} \rightarrow du = \sqrt{a} dv, \quad w = \sqrt{av} + \sqrt{b} \rightarrow dw = \sqrt{a} dv$$

$$\frac{1}{2\sqrt{b}} \left(\frac{1}{\sqrt{a}} \left(\int \frac{du}{u} - \int \frac{dw}{w} \right) \right) = \frac{1}{2\sqrt{ab}} (\ln(|u|) - \ln(|w|))$$

$$= \frac{1}{2\sqrt{ab}} (\ln(|\sqrt{av} - \sqrt{b}|) - \ln(|\sqrt{av} + \sqrt{b}|)) = t + c$$

$$\frac{\ln \left(\left| \frac{\sqrt{av} - \sqrt{b}}{\sqrt{av} + \sqrt{b}} \right| \right)}{2\sqrt{ab}} = \frac{\ln \left(\left| \frac{\sqrt{a}}{\sqrt{a}} * \frac{\sqrt{av} - \sqrt{b}}{\sqrt{av} + \sqrt{b}} \right| \right)}{2\sqrt{ab}} = t + c$$

$$\frac{\ln \left(\left| \frac{av - \sqrt{ab}}{av + \sqrt{ab}} \right| \right)}{2\sqrt{ab}} = t + c$$

Where c is a constant. When t = 0, v = 0, since the plane hasn't started moving, meaning

$$\frac{\ln \left(\left| \frac{a(0) - \sqrt{ab}}{a(0) + \sqrt{ab}} \right| \right)}{2\sqrt{ab}} = (0) + c$$

$$\frac{\ln(1)}{2\sqrt{ab}} = c \rightarrow c = 0 \rightarrow \frac{\ln \left(\left| \frac{av - \sqrt{ab}}{av + \sqrt{ab}} \right| \right)}{2\sqrt{ab}} = t$$

$$\ln \left(\left| \frac{av - \sqrt{ab}}{av + \sqrt{ab}} \right| \right) = 2t\sqrt{ab}$$

$$\frac{av - \sqrt{ab}}{av + \sqrt{ab}} = -e^{2t\sqrt{ab}}$$

$$av - \sqrt{ab} = -ave^{2t\sqrt{ab}} - e^{2t\sqrt{ab}}\sqrt{ab}$$

$$av + ave^{2t\sqrt{ab}} = -e^{2t\sqrt{ab}}\sqrt{ab} + \sqrt{ab}$$

$$(a + ae^{2t\sqrt{ab}})v = -e^{2t\sqrt{ab}}\sqrt{ab} + \sqrt{ab}$$

$$v = \frac{-e^{2t\sqrt{ab}}\sqrt{ab} + \sqrt{ab}}{a + ae^{2t\sqrt{ab}}}$$

Now that we have a function for the velocity of the aircraft dependent only on the time passed, we can do a definite interval from 0 to the total time it's going to be running, t_{empty} hours. Since distance is the antiderivative of velocity, using the integral will just leave us with distance as a function of time. To find out when the gas tank is going to empty, AKA upper bound of the integral, we divide the mass of gasoline on board by the consumption rate, C_R , found above to be $0.004409 \frac{kg}{s}$

$$t_{empty} = \frac{tank_{full\ weight}}{C_R} = \frac{144kg}{0.004409 \frac{kg}{s}} = 32659.2\ s = 9.072\ hours$$

$$v = \frac{-e^{2t\sqrt{ab}}\sqrt{ab} + \sqrt{ab}}{a + ae^{2t\sqrt{ab}}}$$

$$\int_0^{t_{empty}} v\ dt = \int_0^{t_{empty}} \frac{-e^{2t\sqrt{ab}}\sqrt{ab} + \sqrt{ab}}{a + ae^{2t\sqrt{ab}}} dt$$

$$d_{t_{empty}} - d_{t_0} = -\sqrt{\frac{b}{a}} \int_0^{t_{empty}} \frac{e^{2t\sqrt{ab}} - 1}{e^{2t\sqrt{ab}} + 1} dt, \quad u = -2t\sqrt{ab} \rightarrow du = -2\sqrt{ab}\ dt$$

$$d_{total} = -\sqrt{\frac{b}{a}} \left(-\frac{1}{2\sqrt{ab}} \right) \int_0^{-2t_{empty}\sqrt{ab}} \frac{e^{-u} - 1}{e^{-u} + 1} du, \quad w = -u \rightarrow dw = -du$$

$$d_{total} = -\frac{1}{2a} \int_0^{2t_{empty}\sqrt{ab}} \frac{e^w - 1}{e^w + 1} dw$$

$$d_{total} = -\frac{1}{2a} \int_0^{2t_{empty}\sqrt{ab}} \frac{e^w + 1 - 2}{e^w + 1} dw = \frac{1}{2a} \int_0^{2t_{empty}\sqrt{ab}} \left(\frac{e^w + 1}{e^w + 1} - \frac{2}{e^w + 1} \right) dw$$

$$d_{total} = -\frac{1}{2a} \left(\int_0^{2t_{empty}\sqrt{ab}} 1 dw - \int_0^{2t_{empty}\sqrt{ab}} \frac{2}{e^w + 1} dw \right) \quad r = e^w + 1 \rightarrow dr = e^w dw$$

$$d_{total} = -\frac{1}{2a} \left(2t_{empty}\sqrt{ab} - 2 \int_2^{e^{2t_{empty}\sqrt{ab}}+1} \frac{1}{r(r-1)} dr \right)$$

$$= -\frac{1}{2a} \left(2t_{empty}\sqrt{ab} - 2 \int_2^{e^{2t_{empty}\sqrt{ab}}+1} \left(\frac{A}{r} + \frac{B}{r-1} \right) dr \right)$$

$$\rightarrow A(r-1) + Br = 1$$

$$\text{When } r = 1, A(0) + B(1) = 1 \rightarrow B = 1$$

$$\text{When } r = 0, A(0-1) + B(0) = 1 \rightarrow A = -1$$

$$d_{total} = -\frac{1}{2a} \left(2t_{empty}\sqrt{ab} - 2 \int_2^{e^{2t_{empty}\sqrt{ab}}+1} \left(\frac{1}{r-1} - \frac{1}{r} \right) dr \right)$$

$$d_{total} = -t_{empty} \sqrt{\frac{b}{a}} + \frac{1}{a} \left(\ln(|r-1|) \Big|_2^{e^{2t_{empty}\sqrt{ab}}+1} - \ln(|r|) \Big|_2^{e^{2t_{empty}\sqrt{ab}}+1} \right)$$

$$d_{total} = -t_{empty} \sqrt{\frac{b}{a}} + \frac{1}{a} \left(\left(\ln(e^{2t_{empty}\sqrt{ab}} + 1 - 1) - \ln(2 - 1) \right) - \left(\ln(e^{2t_{empty}\sqrt{ab}} + 1) - \ln(2) \right) \right)$$

$$d_{total} = \left(\frac{a}{a} * -t_{empty} \sqrt{\frac{b}{a}} \right) + \frac{1}{a} \left(\ln(e^{2t_{empty}\sqrt{ab}}) - \ln(e^{2t_{empty}\sqrt{ab}} + 1) + \ln(2) \right)$$

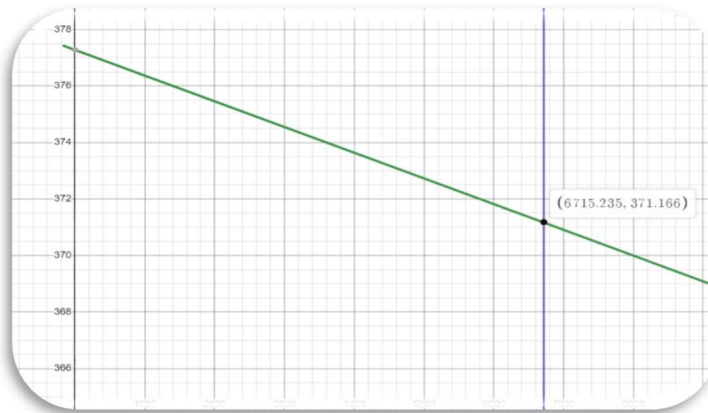
$$d_{total} = \frac{2t_{empty}\sqrt{ab} - t_{empty}\sqrt{ab} - \ln(e^{2t_{empty}\sqrt{ab}} + 1) + \ln(2)}{a}$$

$$d_{total} = \frac{t_{empty}\sqrt{ab} - \ln(e^{2t_{empty}\sqrt{ab}} + 1) + \ln(2)}{a}$$

We can now also decompress all of the variables, keeping in mind that $\frac{\rho C_{L_b} S_b - 2C_d \rho S_{ref}}{4(m_{cargo} + m_{plane})} = a$

and $-\frac{\rho C_{L_b} S_b v_{tip}^2}{6(m_{cargo} + m_{plane})} = b$, and that $m_{plane} = 762 + 70 + 144 = 976 \text{ kg}$.

Meaning we now have a function which directly connects the total distance travelled with the mass of the cargo in the plane. Plotting it from $x=0$ to $x=6715.235$ on Desmos, we find:



Where the y-axis is the distance travelled, in kilometers, and the x-axis is the amount of cargo in the plane, in kilograms. As a result, we find that the addition of extra cargo causes minimal changes to the maximum range of the plane, with a difference of only around 7 kilometers. Since the planes only have to accelerate to such a low velocity, the only difference in timing is at takeoff, while the ranges are functionally the same. With that much, the graph tells us that the plane will have a range of 371.166 kilometers after 9.07 hours with full cargo.

Also interesting to note is the necessary runway length for the Cessna 172 to take off with its full fuel weight. First, assuming a load of 75% capacity, the takeoff speed necessary would be:

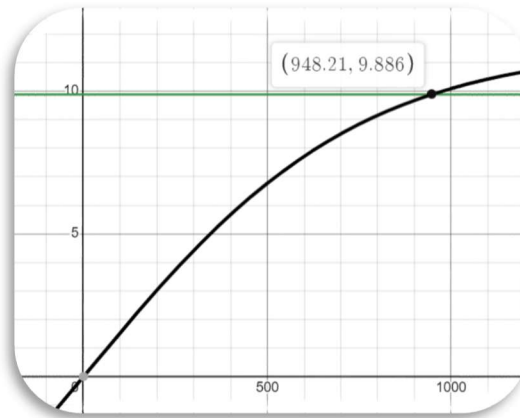
$$Lift = Weight \rightarrow \frac{1}{2} \rho v_{\infty}^2 C_L S_{ref} = mg$$

$$v_{\infty} = \sqrt{\frac{2mg}{\rho C_L S_{ref}}} = \sqrt{\frac{2(7691.235 \text{ kg} * 75\%) \left(1.35 \frac{\text{m}}{\text{s}^2}\right)}{\left(5.4 \frac{\text{kg}}{\text{m}^3}\right)(1.775)(16.165 \text{ m}^2)}} = 10.026 \frac{\text{m}}{\text{s}}$$

We can refer again to the velocity equation,

$$v = \frac{-e^{2t\sqrt{ab}}\sqrt{ab} + \sqrt{ab}}{a + ae^{2t\sqrt{ab}}}$$

Graphing using Desmos, we find that the plane, with a full fuel tank, 75% maximum cargo, and a pilot takes 948.21 seconds to reach lift velocity.



We can then find the length necessary with,

$$\int_0^{948.21} \frac{-e^{2t\sqrt{ab}}\sqrt{ab} + \sqrt{ab}}{a + ae^{2t\sqrt{ab}}} dt = 5643 \text{ m}$$

Due to how much the propeller is limited in speed, the acceleration of the Cessna is pitiful, meaning there needs to be more than 11 times the takeoff distance of the Cessna 172 on Earth, standing at 497 m (“Cessna Skyhawk”)! With this in mind, one can envision a system of transportation across Titan, made up of runways across the equator. With a diameter of 5150 km, Titan’s circumference is $5150\pi \text{ km} = 16179.202 \text{ km}$ (Lorentz et al 375). With a travel distance

of 371.166 km for each Cessna, we would need a fleet of $\frac{16179.202 \text{ km}}{371.166 \text{ km}} = 43.59 \text{ planes} = 44 \text{ planes}$. With a flight time of 9.072 hours each, that gives a total round-trip time of

$$44 \text{ planes} * 9.072 \text{ hours} = 399.771 \text{ hours} = 16.657 \text{ days}$$

As a result, we can see that a Cessna-based transportation system of small cargo across Titan would be inefficient and take almost a half month to circumnavigate the planet.

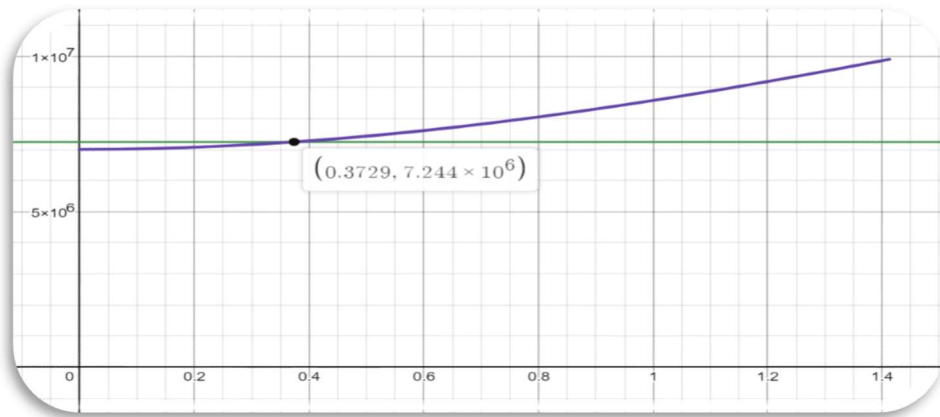
As a result, it is clear that if the Cessna would want to see actual implementation on Titan, it needs to change the design of its propeller to allow for higher speeds, and therefore longer ranges, without destabilizing the airplane. Calling back to the Reynolds Number equation for the propellers, and remembering that the propellers can only handle up to a number of $7.2 * 10^6$:

$$Re_b = \frac{\rho v_{\infty} c}{\mu} \sqrt{1 + \frac{1}{3} \left(\frac{\sqrt{6 \left(\frac{C_d v_{\infty}^2 S_{ref}}{2 C_{L_b} S_b} - \frac{1}{4} v_{\infty}^2 \right)}}{v_{\infty}} \right)^2}$$

Realize that both the chord length, c , equal to 0.577 times the blade length, and the surface area of the blades, S_b , are dependent on the length of the propellers (Burgers 10). As a result, I will replace the chord length with $c = 0.577 radius_{blade}$ and the surface area with $S_b = 0.208 * \left(\frac{radius_{blade}}{0.9525 \text{ m}} \right)^2$, using the square cube law to show that the surface area of the blades decreases proportionally to the square of how much the radius was reduced (“The Square-Cube Law”). Plugging these in, as well as plugging in our new maximum velocity, we get the following equation for the Reynolds Number as a function of the blade radius:

$$Re_b = \frac{\rho v_\infty (0.577 r_{blade})}{\mu} \sqrt{1 + \frac{1}{3} \left(\frac{\sqrt{6 \left(\frac{C_d v_\infty^2 S_{ref}}{2 C_{L_b} \left(0.208 \left(\frac{r_{blade}}{0.9525} \right)^2} - \frac{1}{4} v_\infty^2 \right)} \right)^2}}{v_\infty} \right)^2}$$

Now, we just have to choose a maximum velocity. If we instead limit the maximum velocity of the plane to 13.5 m/s using Desmos:



We are able to find a configuration where the propeller blades have a radius of 0.3729 m. As a result, we have developed a Titan-model Cessna 172, which has propellers almost 2.6 times shorter to profit off of atmospheric conditions. This plane would be able to travel 16.6% faster than the regular Cessna 172 at 13.5 m/s rather than 11.577 m/s and would be specifically designed for the Titan environment.

In conclusion, while at first Titan's low gravity and high atmospheric density seemed to provide the ideal conditions for propelled flight, the reality of flying on Titan with a propeller-based commercial aircraft like the Cessna 172 is much different. By far the biggest limiting

factor is the instability of the propellers at such a low viscosity atmosphere, severely limiting their maximum RPM as to not destabilize the aircraft, and leaving an small difference between the minimum speed required to take off and the maximum speed the plane can go without destabilizing. Flights on the Cessna 172 in Titan would be slow at around 11 m/s, long, cramped, and dangerous as the difference between a slow ascent and slowly plummeting to the ground is found within a few meters per second of each other. Overall, were Cessna 172s to be implemented in Titan, they could be used as cheap, yet inefficient and medium-speed cargo planes, able to transport medium to large amounts of cargo of up to 7700 kg for medium distances, while requiring short infrastructure in the form of long 6000 m runways. However, fuel consumption would be high and inefficient for the amount of work being done, and if the Cessna would want to see actual implementation on Titan, it needs to change the design of its propeller to allow for higher speeds without destabilizing the airplane.

Overall, this internal assessment has opened my eyes to the design process aerospace engineers undergo in designing the specifications of aircraft. So much has to be finely tuned to ensure its stability, the maximization of its distance and velocity, and its complete adaptation to its climate. As a result, it should not be surprising that the Cessna 172 suffered once introduced to Titan's completely new parameters. It would take another round of designing, reducing the propeller length, making the aircraft lighter, and increasing wingspan to make the ideal aerial propeller vessel on Titan. I now have a newfound appreciation for all the variables and considerations engineers must go through to ensure the safety of their creations and the people operating them. I also now understand more fully the mathematics involved and the differential equations of ever-changing accelerations and velocities, only adding to my enthusiasm.

Works Cited

“Air Mass / Density.” *EarthData*, NASA,

<https://www.earthdata.nasa.gov/topics/atmosphere/atmospheric-pressure/air-mass-density#:~:text=Pure%2C%20dry%20air%20has%20a,of%20at%20least%2050%20km..>

Accessed 29 Jan. 2024.

Burgers, Phillip. “A thrust equation treats propellers and rotors as aerodynamic cycles and calculates their thrust without resorting to the blade element method.” *International Journal of Aeronautics and Aerospace*, Embry Riddle Aeronautical University, vol. 6, no. 5, 2019, pp. 1-20,

<https://commons.erau.edu/cgi/viewcontent.cgi?article=1427&context=ijaaa>. Accessed 29

Jan. 2024.

“Cessna Skyhawk.” *Cessna*, Tetron Aviation, <https://cessna.txtav.com/en/piston/cessna-skyhawk>.

Accessed 29 Jan. 2024.

Draley, Eugene C. “Propellers at High Speeds.” Langley Memorial Aeronautical Laboratory, pp. 67-74, <https://ntrs.nasa.gov/api/citations/19790079928/downloads/19790079928.pdf>.

Accessed 29 Jan 2024.

Hall, Nancy. “Four Forces on an Airplane.” *Glenn Research Center*, NASA, 21 Jul. 2022,

<https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/four-forces-on-an-airplane/>.

Accessed 29 Jan. 2024.

Hall, Nancy. “Modern Drag Equation.” *Glenn Research Center*, NASA, 21 Jan. 2023,

<https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/modern-drag-equation/>.

Accessed 29 Jan. 2024.

- Hall, Nancy. "Weight Equation." *Glenn Research Center*, NASA, 29 Jul. 2022, <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/weight-equation/>. Accessed 23 Feb. 2024.
- Hautaluoma, Grey and Alana Johnson. "NASA's Dragonfly Will Fly Around Titan Looking for Origins, Signs of Life," *NASA*, NASA, 27 Jun. 2019, <https://www.nasa.gov/news-release/nasas-dragonfly-will-fly-around-titan-looking-for-origins-signs-of-life/>. Accessed 12 Dec. 2023.
- Lorenz, Ralph D., et al. "Dragonfly: A Rotorcraft Lander Concept for Scientific Exploration at Titan." *John Hopkins APL Technical Digest*, NASA, vol. 34, no. 3, 2018, pp. 374-383, https://dragonfly.jhuapl.edu/News-and-Resources/docs/34_03-Lorenz.pdf. Accessed 29 Jan. 2024.
- "Mars Helicopter," *Mars Communication Team*, NASA, <https://mars.nasa.gov/technology/helicopter/#>. Accessed 12 Dec. 2023.
- McIver, John. "Cessna Skyhawk II / 100 Performance Assessment." *Temporal Images*, pp. 1-19, <http://www.temporal.com.au/c172.pdf>. Accessed 29 Jan. 2023.
- O'Hare, Maureen. "The world's best-selling airplanes." *Cnn travel*, CNN, 28 Jan. 2022, <https://www.cnn.com/travel/article/world-best-selling-airplanes/index.html>. Accessed 29 Jan. 2024.
- "Operator's Manual Lycoming." *Lycoming*, Lycoming, Oct. 2005, 8th ed, pp. 324, <https://www.lycoming.com/sites/default/files/attachments/O-HO-IO-HIO-AIO%2520%2526%2520TIO-360%2520Oper%2520Manual%252060297-12.pdf>. Accessed 29 Jan. 2024.

“Reynolds Number.” *Glenn Research Center*, NASA, <https://www.grc.nasa.gov/www/k-12/airplane/reynolds.html>. Accessed 29 Jan. 2024.

“The Square-Cube Law in Chemistry and Biology.” *Drug Hunter*, Drug Hunter Inc, 19 Jan. 2021, <https://drughunter.com/articles/sphericalcow/>. Accessed 23 Feb. 2024.

Tavernier, Lyle. “May the Force = mass x acceleration.” *Jet Propulsion Laboratory*, California Institute of Technology, 28 Apr. 2023, <https://www.jpl.nasa.gov/edu/news/2016/05/03/may-the-force-equals-mass-x-acceleration/>. Accessed 23 Feb. 2024.

“Titan: Facts,” *NASA*, NASA, <https://science.nasa.gov/saturn/moons/titan/facts/>. Accessed 12 Dec. 2023.